

MMIS 885: Management Information System

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Lecture 2:MMIS 885

Information Technology (IT)
Infrastructure

Lecture 2: Outline

- Introduction
- Evolution of IT infrastructure
- Key drivers of IT infrastructure Evolution
- IT infrastructure components
- Contemporary Hardware platform trend
- Contemporary Software platform Trend
- Management Issues

What is IT Infrastructure?

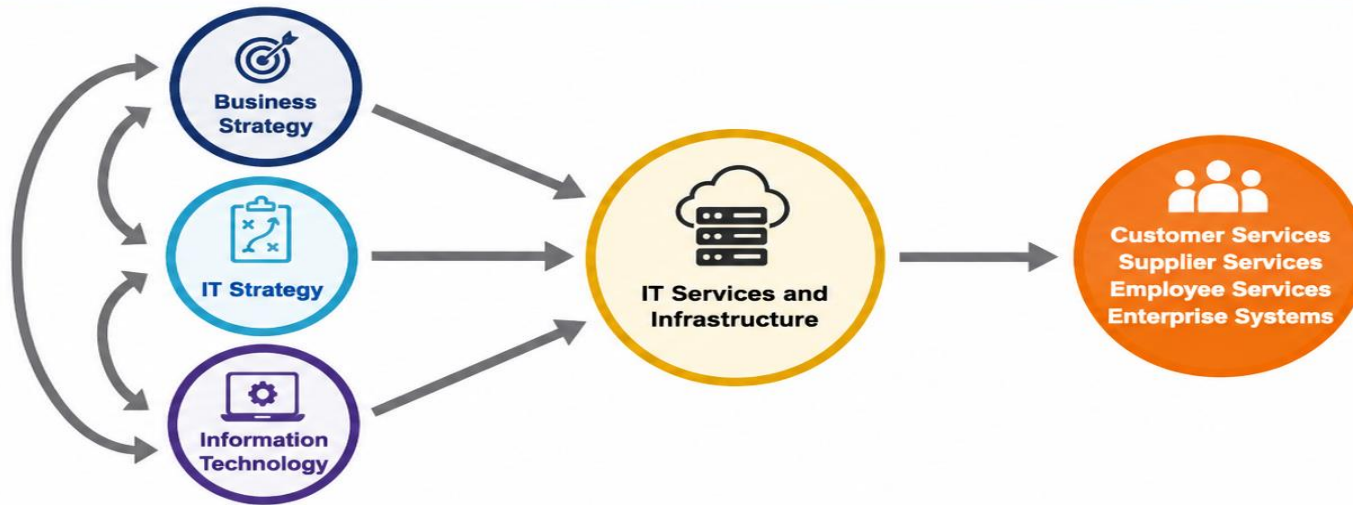
- **Information Technology (IT) Infrastructure** is the backbone of modern organizations.
- **IT infrastructure**
 - **set of** Physical and virtual resources required to run an organization
 - collection of IT resources and services that support the entire organization including:
 - Computing platforms providing computing services
 - Telecommunications services
 - Data management services
 - Application software services
 - Etc.
- **IT infrastructure** enables operational efficiency, innovation, competitive advantage, and informed decision-making

Connection between Firm, IT infrastructure and Business Capabilities



Management Information Systems Chapter 5: IT Infrastructure and Emerging Technologies

CONNECTION BETWEEN THE FIRM, IT INFRASTRUCTURE, AND BUSINESS CAPABILITIES



What this means: The services a firm provides to its customers, suppliers, and employees depend directly on its IT infrastructure. Ideally, this infrastructure should support the firm's **business and information systems strategy**. New information technologies have a **powerful impact** on both **business** and **IT strategies**, as well as the **services** that can be delivered.



Business Strategy

What the business wants to achieve



IT Strategy

How IT will help the business succeed



Information Technology

The hardware, software, data, and networks used



IT Services and Infrastructure

The foundation that delivers IT services to the organization



Business Capabilities (Services)

Value delivered to customers, suppliers, employees

Evolution of IT infrastructure

- **General-purpose mainframe and minicomputer era: 1959 to present**
 - 1958: IBM first mainframes introduced
 - 1965: less expensive DEC minicomputers introduced
- **– Personal computer era: 1981 to present**
 - 1981: Introduction of IBM PC
 - Proliferation in 80s, 90s resulted in growth of personal software
- **– Client/server era: 1983 to present**
 - Desktop clients networked to servers, with processing work split between clients and servers
 - Network may be two-tiered or multitiered (N-tiered)
 - Various types of servers (network, application, Web)

Evolution of IT infrastructure (cont.)

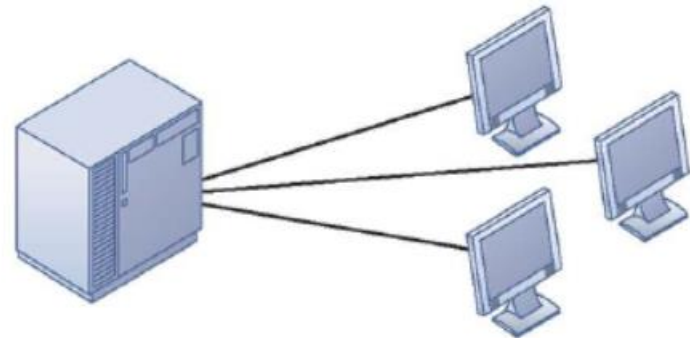
- **Enterprise computing era: 1992 to present**
 - Systems that integrate computers, software, databases, and networks to support the activities and goals of an organization
 - Move toward integrating disparate networks, applications using Internet standards and enterprise applications
- **Cloud and mobile computing: 2000 to present**
 - Cloud computing: computing power and software applications supplied over the Internet or other network
- – Fastest growing form of computing

Evolution of IT infrastructure (cont.)

Illustrated here are the typical computing configurations characterizing each of the five eras of IT infrastructure evolution.

FIGURE 5-2

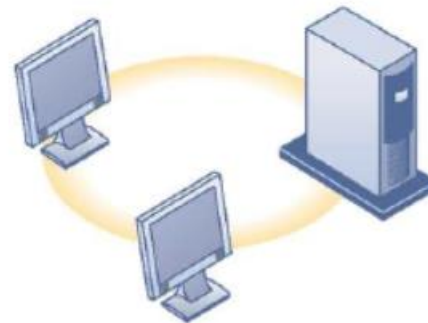
Mainframe/
Minicomputer
(1959–present)



Personal
Computer
(1981–present)



Client/Server
(1983–present)

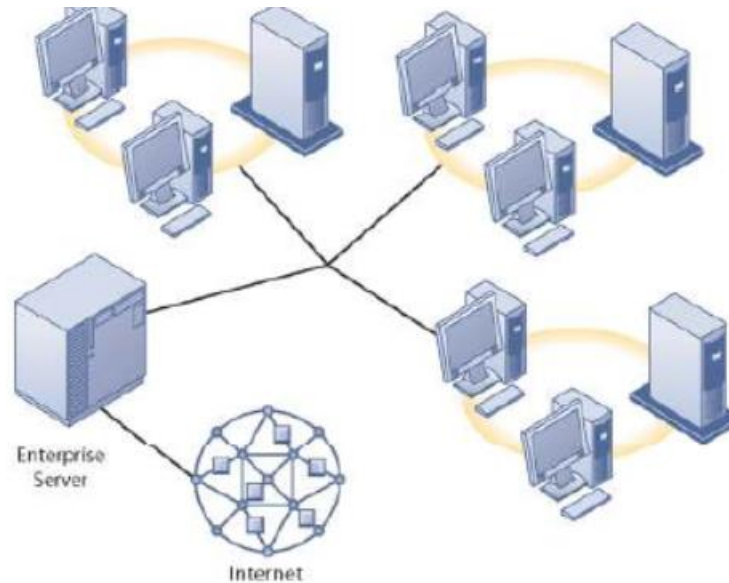


Evolution of IT infrastructure (cont.)

Illustrated here are the typical computing configurations characterizing each of the five eras of IT infrastructure evolution.

FIGURE 5-2

Enterprise
Computing
1992–present)



Cloud and Mobile
Computing
(2000–present)



Technology drivers of infrastructure evolution

- **Moore's law and micro processing power**
 - Computing power doubles every 18 months
 - Nanotechnology
 - Shrinks size of transistors to size comparable to size of a virus
- **Law of Mass Digital Storage**
 - The amount of data being stored each year doubles

MOORE'S LAW AND MICROPROCESSOR PERFORMANCE

Moore's Law and Microprocessor Performance

Over the past 50+ years, microprocessor performance (measured by transistors, clock speed, and cores) has increased exponentially.

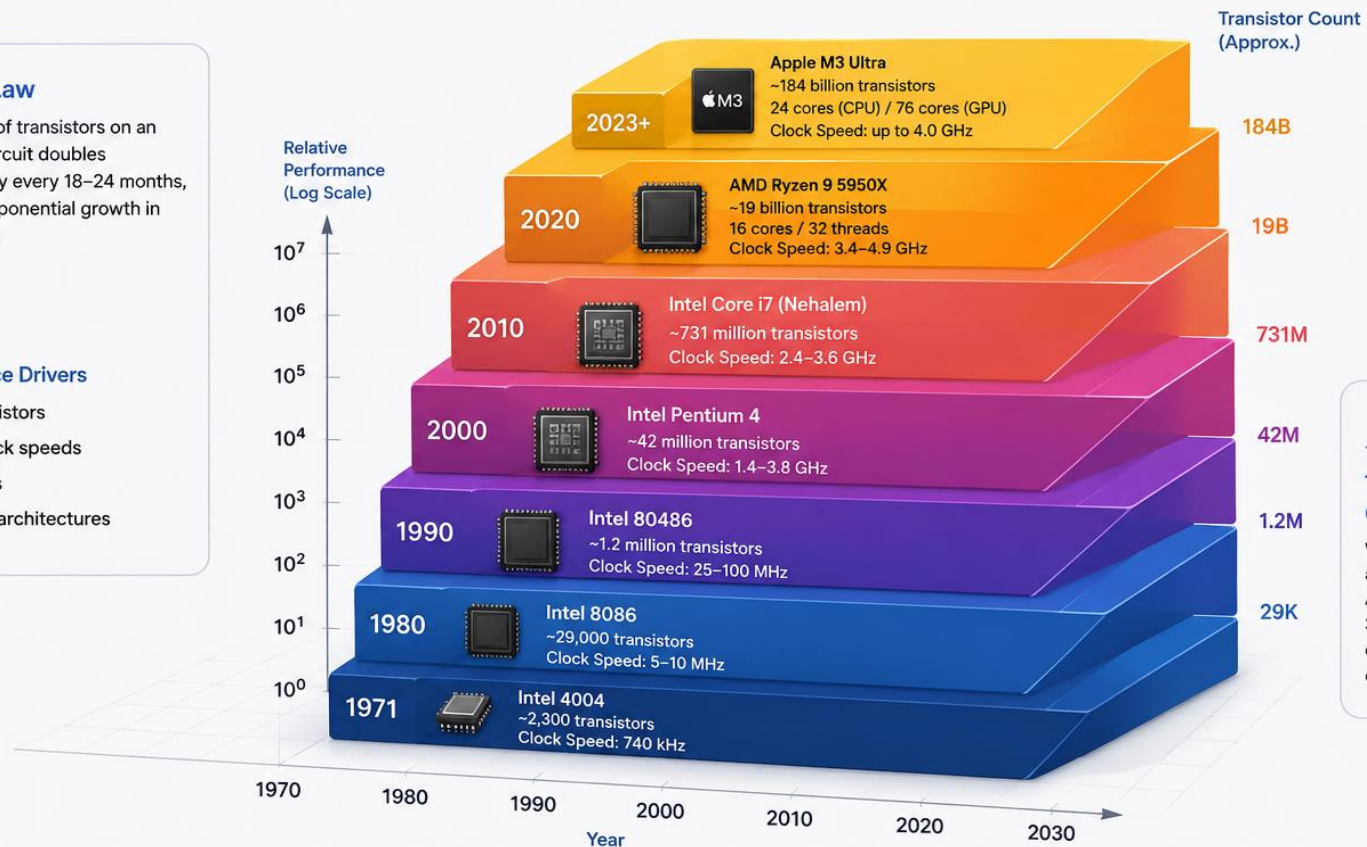
Moore's Law

The number of transistors on an integrated circuit doubles approximately every 18–24 months, leading to exponential growth in performance.



Performance Drivers

- More transistors
- Higher clock speeds
- More cores
- Advanced architectures



The Trend Continues

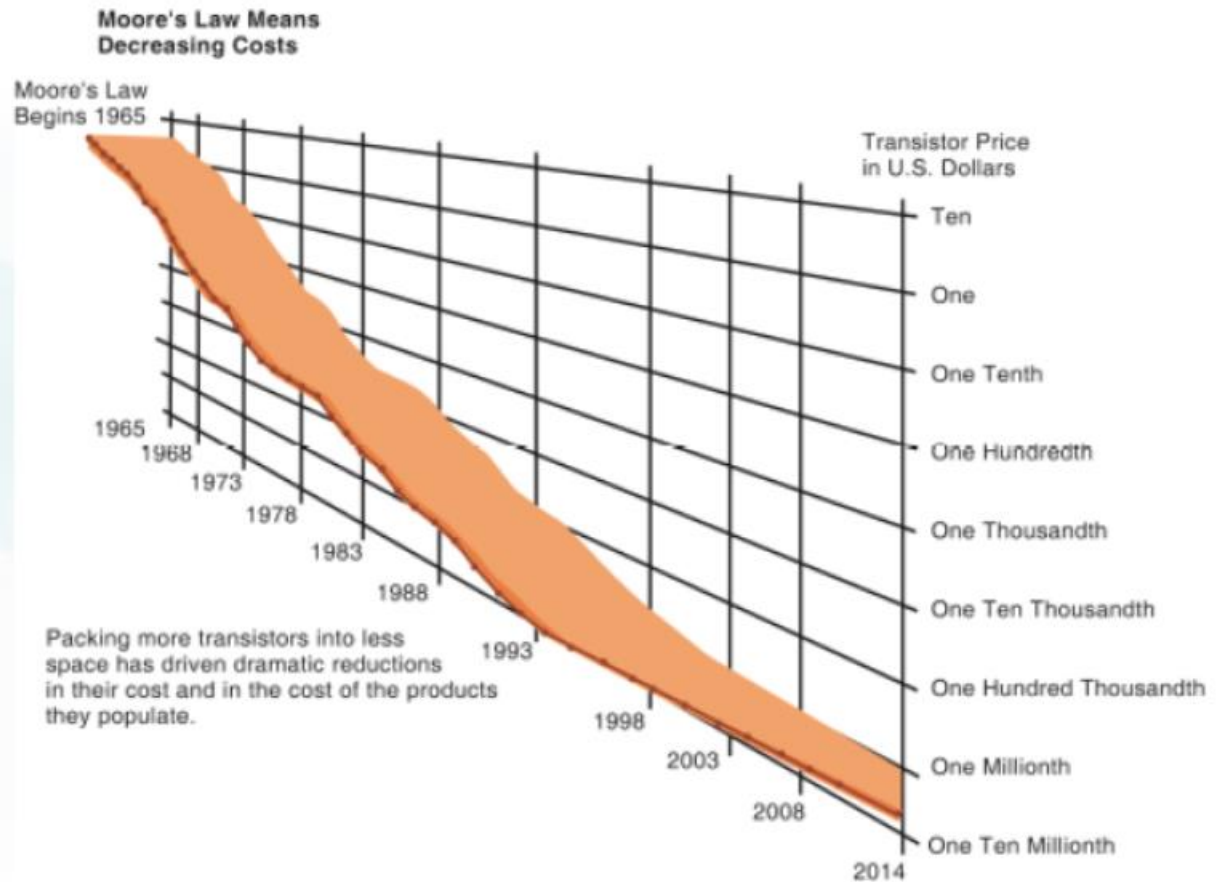
With advanced chip architectures, AI acceleration, and 3D stacking, performance continues to grow exponentially.

Note: Transistor counts are approximate and represent total transistors on the chip.
Sources: Intel, AMD, Apple, IEEE Spectrum, and other public disclosures.

FALLING COST OF CHIPS

Packing more transistors into less space has driven down transistor cost dramatically as well as the cost of the products in which they are used.

FIGURE 5-5



Technology drivers of infrastructure evolution

- **Metcalfe's Law and network economics**
 - Value or power of a network grows exponentially as a function of the number of network members
 - As network members increase, more people want to use it (demand for network access increases)
- **Declining communication costs and the Internet**
 - An estimated 2.3 billion people worldwide have Internet access
 - As communication costs fall toward a very small number and approach 0, utilization of communication and computing facilities explodes

IT infrastructure components

- IT Infrastructure has seven main components
 - 1. Computer hardware platforms
 - 2. Operating system platforms
 - 3. Enterprise software applications
 - 4. Data management and storage
 - 5. Networking/telecommunications platforms
 - 6. Internet platforms
 - 7. Consulting system integration services

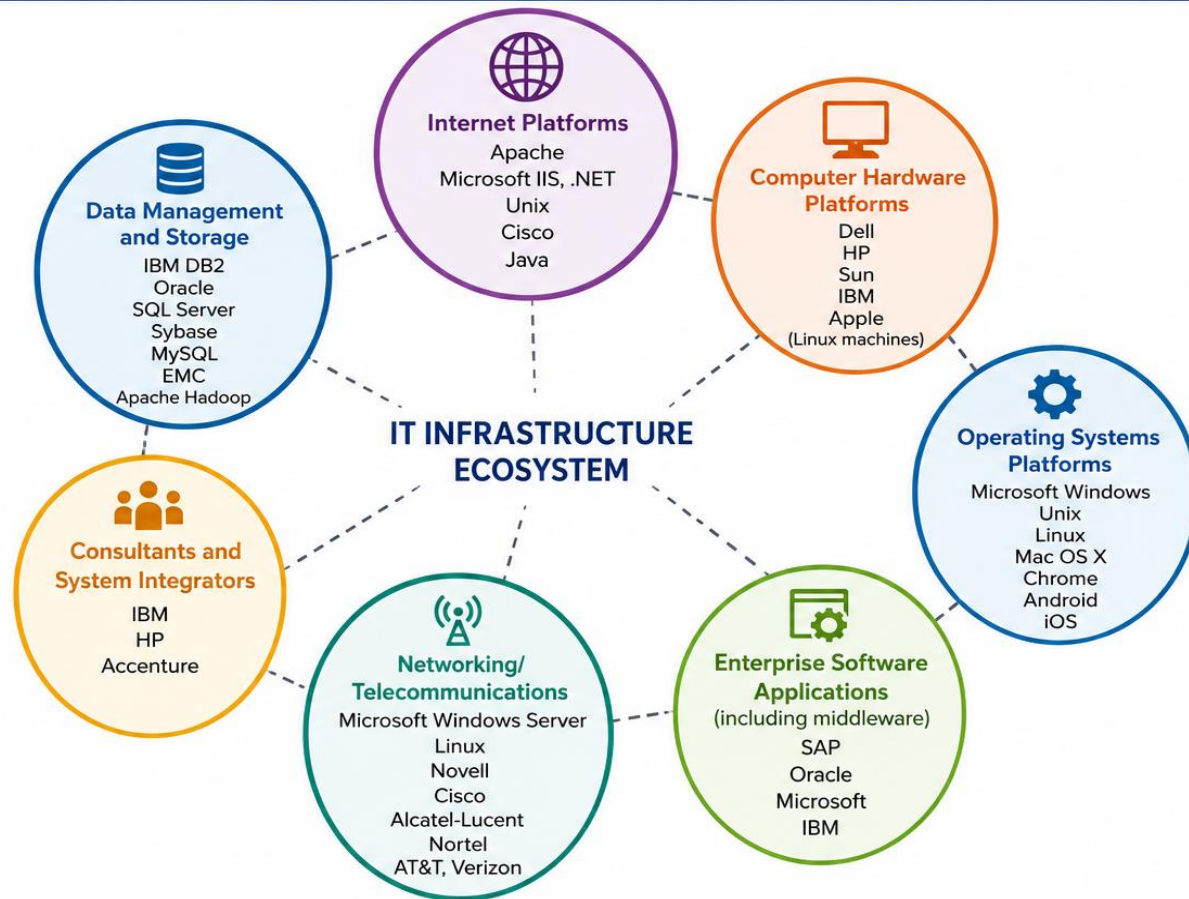
IT infrastructure Ecosystem



THE IT INFRASTRUCTURE ECOSYSTEM

There are seven major components that must be coordinated to provide the firm with a coherent IT infrastructure. Listed here are major technologies and suppliers for each component.

FIGURE 5-9



Computer hardware platforms

- **Physical computing devices such as:**
- **Client machines**
 - Desktop PCs, mobile devices—PDAs, laptops
- **Servers**
 - Blade servers: ultrathin computers stored in racks
- **Mainframes**
 - IBM mainframe equivalent to thousands of blade servers
- **Top chip producers: AMD, Intel, IBM**
- **Top firms: IBM, HP, Dell, Sun Microsystems**

Operating system platforms

- **Systems responsible for running and managing the IT infrastructure**
 - Server level:
 - 65% run Unix or Linux; 35% run Windows
 - Client level:
 - 90% run Microsoft Windows (XP, 2000, CE, etc.)
- – Mobile/multitouch (Android, iOS)
- – Cloud computing (Google's Chrome OS)

Enterprise software applications

- **Software that supports organizational operations and decision-making**
 - **Enterprise-wide applications**
 - Enterprise Resource Planning (ERP)
 - Customer Relationship Management (CRM)
 - Supply Chain Management (SCM)
 - Human Resource Management Systems (HRMS)
 - **Business Intelligence & Analytics**
 - Data visualization and reporting tools
 - AI-powered analytics and decision support
 - **Collaboration & Productivity Platforms**
 - Microsoft 365, Google Workspace
 - Project and workflow management tools
 - **Leading Enterprise Software Vendors**
 - SAP
 - Oracle
 - Microsoft
 - Salesforce
 - Workday
 - ServiceNow

Data management and storage

- **Technologies used to store, organize, process, and protect organizational data**
 - **Database Management Systems (DBMS)**
 - Relational databases (MySQL, Oracle, SQL Server, PostgreSQL)
 - NoSQL databases (MongoDB, Cassandra)
 - **Enterprise Storage Solutions**
 - Storage Area Networks (SAN)
 - **Modern Data Platforms**
 - Data warehouses
 - Big data platforms
 - **Cloud Storage Providers**
 - Amazon S3
 - Microsoft Azure Storage
 - Google Cloud Storage

Networking/telecommunications platforms

- **Infrastructure that enables communication and data exchange within and between organizations**
 - **Enterprise Networks**
 - Local Area Networks (LAN)
 - Wide Area Networks (WAN)
 - **Wireless Communication Technologies**
 - Wi-Fi 6/6E and Wi-Fi 7
 - 4G LTE and 5G mobile networks
 - **Communication Services**
 - Voice over IP (VoIP)
 - Video conferencing systems
 - Unified communications platforms
 - **Network Equipment Providers**
 - Cisco Systems
 - Huawei

Internet platforms

- **Technologies and services that enable organizations to operate and deliver services over the Internet**
 - **Web and Application Platforms**
 - Web servers and application servers
 - Content Management Systems (CMS)
 - E-commerce platforms
 - **Cloud-Based Internet Services**
 - Infrastructure as a Service (IaaS)
 - Platform as a Service (PaaS)
 - Software as a Service (SaaS)
 - **Digital Business Technologies**
 - Web portals
 - APIs and web service
 - **Major Internet Platform Providers**
 - Amazon Web Services (AWS)
 - Microsoft Azure
 - Google Cloud Platform (GCP)

Consulting and system integration services

- – Even large firms do not have resources for full
- range of support for new, complex infrastructure
- – Software integration: ensuring new
- infrastructure works with legacy systems
- – Legacy systems: older TPS created for
- mainframes that would be too costly to replace
- or redesign
- – Accenture, IBM Global Services, EDS, Infosys,
- Wipro

Contemporary Hardware Platform Trend

- **The Mobile Digital Platform**
 - **Smartphones (Android and iPhone)**
 - Primary computing devices for billions of users worldwide
 - Support high-speed **5G/5G Advanced** connectivity
 - Enable web browsing, email, instant messaging, video conferencing, mobile banking, e-commerce, cloud computing, and AI-powered applications
 - **Tablets (iPad, Samsung Galaxy Tab, Microsoft Surface Go)**
 - Portable devices for education, business, content creation, and entertainment
 - Support stylus input, multitasking, and cloud-based collaboration
 - Widely used for remote learning and hybrid work environments
 - **Wearable Computing Devices**
 - Smartwatches (Apple Watch, Galaxy Watch)
 - Fitness trackers and health-monitoring devices
 - Provide real-time notifications, health analytics, navigation, and immersive digital experiences

Contemporary Hardware Platform Trend

- **Grid Computing**
 - **Distributed computing using multiple interconnected computers**
 - A computing model in which the processing power of many geographically dispersed computers is combined to solve large and complex computational problems.
 - Resources from multiple computers work together as a "virtual supercomputer."
 - **How Grid Computing Works**
 - Large tasks are divided into smaller subtasks.
 - Subtasks are distributed across multiple computers (nodes).
 - Each node processes its assigned task.
 - Results are collected and combined to produce the final output.
 - **Applications in Organizations**
 - Scientific research and simulations.
 - Weather forecasting and climate modeling.
 - Financial risk analysis.
 - Engineering design and testing.
 - Big data processing and analytics.

Contemporary Hardware Platform Trend

- **Key Benefits**

- Utilizes idle computing resources efficiently.
- Cost-effective alternative to supercomputers.
- High computational power.
- Scalability by adding more nodes.
- Improved resource sharing across organizations.

- **Drawbacks**

- Complex management and coordination.
- Network dependency and communication delays.
- Security and data privacy concerns.
- Performance may vary due to heterogeneous systems.
- Difficult troubleshooting and maintenance.

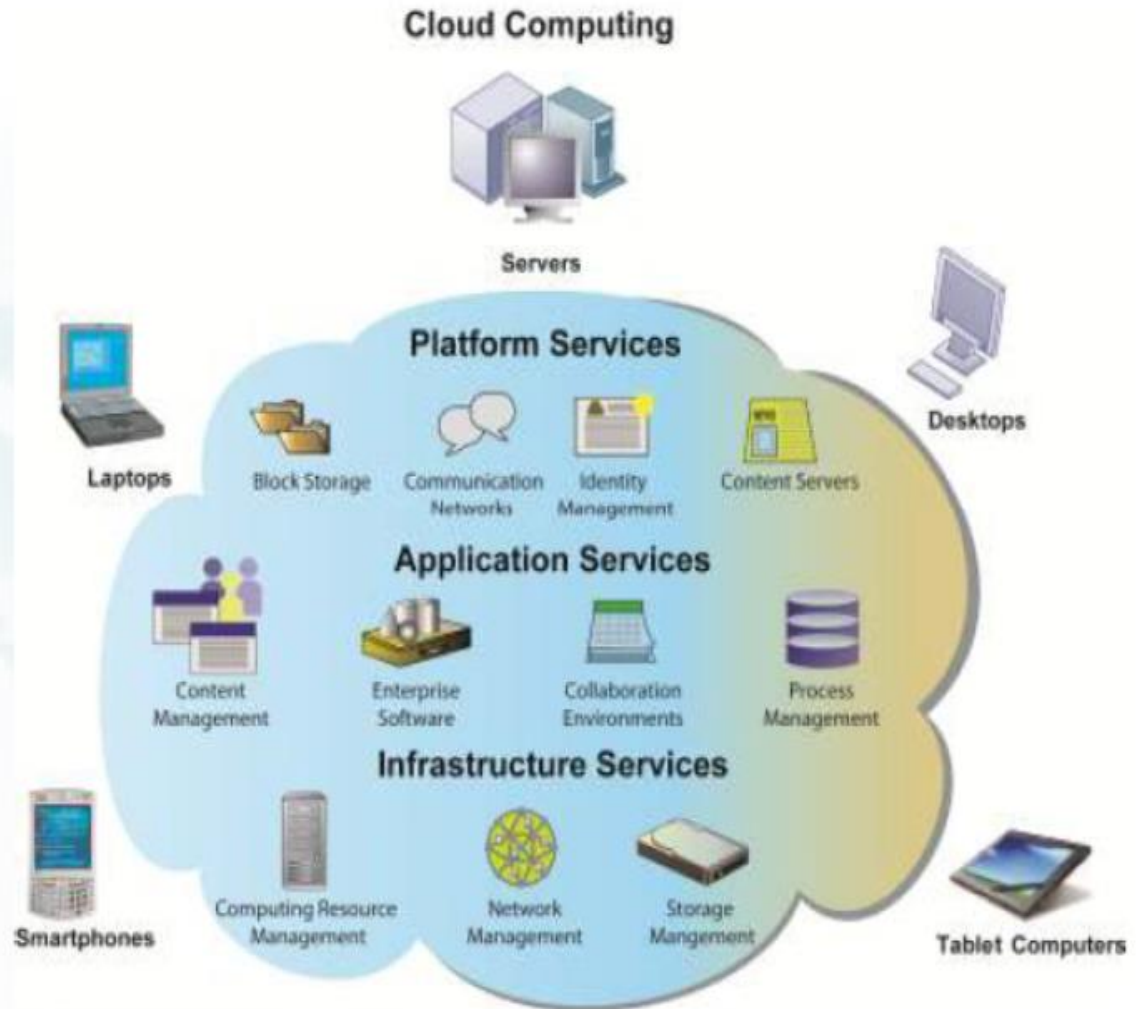
Contemporary Hardware Platform Trend

- **Cloud computing**
 - **On-demand (utility) computing services obtained over network**
 - **Instead of owning hardware/software**, organizations rent resources from cloud providers on a pay-as-you-go basis
 - Cloud Computing service providers provides some of the following:
 - Infrastructure as a service
 - Platform as a service
 - Software as a service
- **Deployment Models**
 - Public cloud, Private cloud, Hybrid cloud, Community cloud
- **Key Benefits**
 - Allows companies to minimize IT investments
 - Scalability and elasticity
 - Disaster recovery capabilities
 - Global accessibility
- **Drawbacks**
 - **Concerns of security and reliability**

Contemporary Hardware Platform Trend

In cloud computing, hardware and software capabilities are a pool of virtualized resources provided over a network, often the Internet. Businesses and employees have access to applications and IT infrastructure anywhere, at any time, and on any device.

Figure 5-10



Contemporary Hardware Platform Trend

- **Virtualization**
 - **Technology that creates virtual versions of computing resources**
 - Virtualization allows a single physical computer or server to run multiple virtual machines (VMs), each functioning as an independent computer with its own operating system and applications.
 - It separates software from physical hardware, enabling more efficient use of computing resources.
 - **Types of Virtualization**
 - **Server Virtualization:** Multiple virtual servers run on a single physical server.
 - **Storage Virtualization:** Multiple storage devices are combined into a single virtual storage pool.
 - **Network Virtualization:** Creates virtual networks independent of physical infrastructure.
 - **How Virtualization Works**
 - A software layer called a **Hypervisor** (or Virtual Machine Monitor) manages virtual machines.
 - The hypervisor allocates CPU, memory, storage, and network resources to each VM.
 - Multiple operating systems can run simultaneously on the same physical hardware.
 - **Common Virtualization Platforms**
 - VMware vSphere , Microsoft Hyper-V , Oracle VirtualBox

Contemporary Hardware Platform Trend

- **Applications in Organizations**
 - Data center consolidation.
 - Cloud computing infrastructure.
 - Software development and testing environments.
 - Disaster recovery and business continuity.
 - Virtual desktop infrastructure (VDI).
 - Hosting multiple applications on a single server.
- **Key Benefits**
 - Reduces hardware acquisition and maintenance costs.
 - Improves resource utilization and efficiency.
 - Simplifies backup and disaster recovery.
 - Supports scalability and flexibility.
 - Reduces energy consumption and data center space requirements.
- **Drawbacks**
 - Initial setup can be complex.
 - Performance overhead may occur in some environments.
 - A failure of the host server can affect multiple virtual machines.
 - Requires skilled personnel for management and maintenance.
 - Licensing costs for some virtualization platforms can be expensive.

Contemporary Hardware Platform Trend

- Virtualization VS. Cloud Computing
- Virtualization is often considered the foundation of cloud computing.
- Cloud providers use virtualization technology to create and manage large pools of computing resources that can be delivered to users on demand.

Feature	Virtualization	Cloud Computing
Definition	Creates virtual versions of hardware and software resources	Delivers computing resources as services over a network
Ownership	Usually managed by the organization	Managed by a cloud provider or organization
Purpose	Improve hardware utilization	Provide on-demand access to computing resources
Infrastructure	Runs on physical servers using hypervisors	Often built using virtualization technologies
Cost Model	Capital expenditure (buying hardware)	Pay-as-you-go or subscription-based
Example	VMware, Hyper-V, VirtualBox	AWS, Microsoft Azure, Google Cloud

Contemporary Hardware Platform Trend

- **Green Computing (Green IT)**
- **Focuses on environmentally sustainable use of information technology**
 - The design, manufacture, use, and disposal of computers, servers, and related systems in ways that reduce environmental impact.
 - Focuses on minimizing energy consumption, reducing electronic waste (e-waste), and promoting sustainability.
- **Key Practices**
 - Use of energy-efficient hardware and devices.
 - Server virtualization and consolidation.
 - Power management techniques (sleep mode, automatic shutdown).
 - Recycling and proper disposal of electronic equipment.
 - Adoption of renewable energy sources for data centers.

Contemporary Hardware Platform Trend

- **Applications in Organizations**
 - Green data centers.
 - Energy-efficient networks and computing infrastructure.
 - Sustainable IT procurement policies.
- **Key Benefits**
 - Reduces energy consumption and operational costs.
 - Minimizes carbon footprint and environmental impact.
 - Extends the lifespan of IT equipment.
 - Helps organizations comply with environmental regulations.
- **Drawbacks**
 - High initial investment for energy-efficient technologies.
 - Cost of replacing legacy systems.
 - Requires employee awareness and organizational commitment.

Contemporary Hardware Platform Trend

- **Autonomic Computing**
- **Self-managing computing systems**
 - Computing systems capable of managing themselves with minimal human intervention.
 - Inspired by the human autonomic nervous system, which automatically regulates body functions such as breathing and heartbeat.
- *Core Characteristics (Self- Properties)**
 - **Self-Configuration:** Automatically configures and adapts to changing environments.
 - **Self-Healing:** Detects and recovers from faults automatically.
 - **Self-Optimization:** Continuously improves performance and resource utilization.
 - **Self-Protection:** Identifies and protects against security threats and attacks.
- **Applications in Organizations**
 - Automated server management.
 - Cloud infrastructure management.
 - Network monitoring and optimization.
 - Cybersecurity and threat detection systems.
 - Intelligent database administration.

Contemporary Hardware Platform Trend

- **Key Benefits**

- Reduces system administration workload.
- Improves system reliability and availability.
- Faster detection and resolution of faults.
- Enhances security through automated monitoring.
- Reduces operational costs.

- **Drawbacks**

- Complex to design and implement.
- High deployment and maintenance costs.
- Reduced human control over decision-making.
- Incorrect automated decisions may affect operations.

Contemporary Software Platform Trends

- **Open-Source Software (OSS)**
 - Software whose source code is publicly available
 - Can be used, modified, and redistributed by individuals and organizations
 - Developed by global communities and commercial contributors
- **Why Organizations Use Open Source**
 - Reduced software acquisition costs
 - Faster innovation and customization
 - Enhanced security through community review
 - Avoidance of vendor lock-in
- **Major Open-Source Technologies**
 - **Linux** (Operating System)
 - **Apache** and **Nginx** (Web Servers)
 - **MySQL** and **PostgreSQL** (Databases)
 - **Python** and **R** (Programming Languages)
 - **TensorFlow** and **PyTorch** (Artificial Intelligence)

Contemporary Software Platform Trends

- **Web Technologies and Modern Application Development**
 - Web applications have evolved from static pages to intelligent, interactive platforms
 - Support real-time collaboration, multimedia, AI integration, and cloud connectivity
- **Core Web Technologies**
 - **HTML5**
 - Current standard for web content development Supports:
 - Audio and video playback
 - Responsive design
 - Interactive graphics
 - Offline web applications
 - **JavaScript Ecosystem**
 - Primary language for modern web applications
 - Popular frameworks:
 - React , Angular , and Next.js
 - **Java**
 - Platform-independent programming language
 - Widely used for enterprise systems, banking applications, and cloud services

Contemporary Software Platform Trends

- **Software outsourcing and cloud services**
 - **Three external sources for software:**
 - Software packages and enterprise software
 - Software outsourcing
 - Contracting outside firms to develop software
 - Cloud-based software services
 - Software as a service (SaaS)
 - Accessed with Web browser over Internet
 - Service Level Agreements (SLAs): formal agreement with service providers

Contemporary Software Platform Trends

- **Mobile Apps and Cross-Platform Development**
 - **Mobile Applications (Apps)**
 - Software designed for smartphones, tablets, wearables, and IoT devices
 - **Types of Apps**
 - Native Apps, Web Apps , Progressive Web Apps (PWAs), and Cross-Platform Apps
 - **Popular Development Frameworks**
 - Flutter, React Native, and Kotlin Multiplatform
 - **Enterprise Uses**
 - Mobile banking
 - E-commerce
 - Healthcare applications
 - Enterprise productivity tools
 - Learning management systems

Contemporary Software Platform Trends

- **Artificial Intelligence and Intelligent Software Platforms**
 - **AI-Powered Software Systems**
 - AI is becoming a core component of modern enterprise software
- **Applications**
 - Intelligent assistants and chatbots
 - Predictive analytics
 - Fraud detection
 - Recommendation systems
 - Automated customer support
 - Generative AI for content creation
- **Popular AI Platforms**
 - OpenAI, Microsoft Copilot, Google Gemini
 - Anthropic Claude, IBM watsonx

Management Issues

- **Dealing with platform and infrastructure change**
- As firms shrink or grow, IT needs to be flexible and scalable
- Scalability:
 - Ability to expand to serve a larger number of users
- For mobile computing and cloud computing
 - New policies and procedures for managing these new platforms
 - Contractual agreements with firms running clouds and distributing software are required

Management Issues

- **Management and governance**
 - **Who controls IT infrastructure?**
 - **How should IT department be organized?**
 - Centralized
 - Central IT department makes decisions
 - Decentralized
 - Business unit IT departments make own decisions
- **– How are costs allocated between divisions, departments?**

Management Issues

- **Making wise infrastructure investments**
 - **Amount to spend on IT is complex question**
 - Rent vs. buy, cloud computing
 - Outsourcing
 - **Total cost of ownership (TCO) model**
 - Analyzes direct and indirect costs
 - Hardware, software account for only about 20% of TCO
 - Other costs: Installation, training, support, maintenance, infrastructure, downtime, space, and energy
 - **TCO can be reduced**
 - Use of cloud services, greater centralization and standardization of hardware and software resources

Management Issues

- **Competitive forces model for IT infrastructure investment**
 - 1. Market demand for firm's services
 - What customers needs
 - 2. Firm's business strategy
 - What the organization wants to achieve
 - 3. Firm's IT strategy, infrastructure, and cost
 - what the organization can afford
 - 4. Information technology assessment
 - What technology is available
 - 5. Competitor firm services
 - What competitors are doing
 - 6. Competitor firm IT infrastructure investments
 - The kind of IT used by competitors

Management Issues

COMPETITIVE FORCES MODEL FOR IT INFRASTRUCTURE



FIGURE 5-12 There are six factors you can use to answer the question, *"How much should our firm spend on IT infrastructure?"*

Management Issues

- Suppose a hospital wants to implement an Electronic Health Record (EHR) system.
 - **Market demand** – patients want online appointment booking
 - **Business Strategy** – the hospital wants to improve healthcare delivery
 - **Technology Trend** - cloud based HER systems are now available
 - **Competitor Services** – competing hospitals already offer digital services
 - **Competitor investment** – competitors are investing in AI diagnostic tools
 - **IT strategy and cost** – the hospital has limited IT budget

Management Issues

- The competitive force model helps managers determine the appropriate level of IT infrastructure investment by balancing:
 - Organizational goal
 - Market demands
 - Technological advancements
 - Competitive pressure
 - and cost consideration

Review Questions

- Should businesses move to the cloud?
 - 1. What business benefits do cloud computing services provide?
 - What problems do they solve?
 - 2. What are the disadvantages of cloud computing?
 - 3. How do the concepts of capacity planning, scalability, and TCO apply to this case?
 - Apply these concepts both to your business or organization and to subscribers of its services.
 - 4. What kinds of businesses are most likely to benefit from using cloud computing? Why?